

KubeSage AI

AI-Powered Multi-Cluster Kubernetes Troubleshooting Platform

Project Vision

KubeSage AI is a platform that enables users to register Kubernetes clusters, monitor their health, investigate failures, identify root causes using AI, and receive actionable remediation recommendations from a single dashboard.

Problem Statement

Kubernetes troubleshooting often requires engineers to manually inspect pod states, logs, events, deployments, and cluster resources to determine why applications fail. This process is time-consuming and requires deep Kubernetes expertise. KubeSage AI aims to automate investigation and provide intelligent root cause analysis.

Project Objectives

- Allow users to register and manage Kubernetes clusters.
- Provide centralized monitoring for multiple clusters.
- Automatically investigate Kubernetes failures.
- Perform AI-driven root cause analysis (RCA).
- Generate remediation and prevention recommendations.
- Maintain investigation history and incident reports.

Supported Cluster Types (MVP)

- Proxmox Kubernetes Cluster (On-Premise)
- Minikube Cluster (Local Development)
- AWS EKS Cluster (Cloud)

Core Features

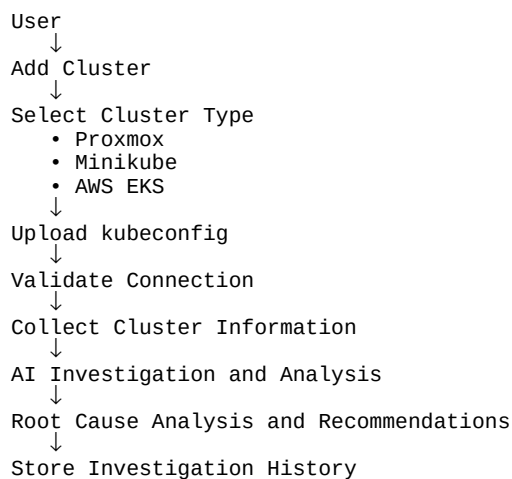
- Cluster Registration and Management
- Multi-Cluster Monitoring Dashboard
- Failure Investigation Engine
- Logs, Events, and Cluster State Analysis
- AI Root Cause Analysis (RCA)
- Suggested Fixes and Remediation Steps
- Prevention Recommendations

- Investigation History and Incident Reports

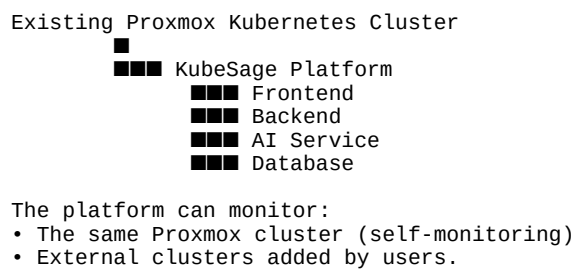
Supported Kubernetes Problems

- CrashLoopBackOff
- ImagePullBackOff
- OOMKilled
- Pending Pods
- Resource Exhaustion
- Deployment Rollout Failures
- Service Selector Mismatch
- DNS Resolution Problems
- Readiness Probe Failures
- Liveness Probe Failures
- Networking Issues

User Workflow

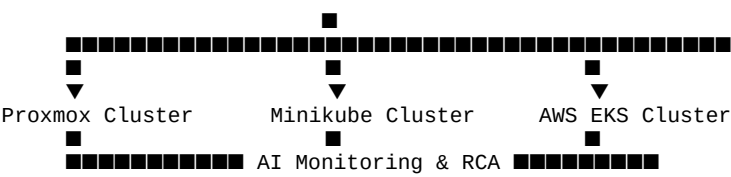


Deployment Architecture



Multi-Cluster Architecture

KubeSage Platform



Tech Stack

Layer	Technology
Frontend	React + TypeScript
Styling	Tailwind CSS
Backend	FastAPI (Python)
Database	PostgreSQL
ORM	SQLAlchemy + Alembic
Authentication	Google OAuth + Email/Password + JWT
Kubernetes Integration	Kubernetes Python Client
AI	OpenAI API
Containerization	Docker
Orchestration	Kubernetes
CI/CD	GitHub Actions
Hosting	Proxmox Kubernetes Cluster